WaferInspectionSuite Setup Guide

Step-by-step instructions to run the repository and local web dashboard on any computer.

Prerequisites

Before running the application, make sure the target computer has the following installed:

- **Java Development Kit (JDK)**: Version 17 or higher recommended.
- **Git**: To clone the repository.

How to Run the Application

Step 1: Clone the Repository

Open a terminal (macOS/Linux) or Command Prompt/Git Bash (Windows) and clone your GitHub repository:

```
git clone https://github.com/USERISNTHERE/WaferInspectionSuite.git
cd WaferInspectionSuite
```

Step 2: Compile the Java Backend

Clean any previous build artifacts, create the target binary directory, and compile all Java source files:

```
rm -rf bin
mkdir bin
javac -d bin $(find src -name "*.java")
```

Step 3: Run the Terminal Robustness & Analysis Suite

Execute the main orchestration class to test multiple manufacturing error profiles (Edge-Ring, Scratch, Center-Cluster) and print executive reports:

```
java -cp bin com.wafer.Main
```

Step 4: Launch and View the Local Web Dashboard

To inspect real-time JSON payloads and view the interactive dashboard:

- Ensure your local server or static preview is running.
- Open your browser and navigate to: `http://localhost:8080`

> **Pro-Tip for Sharing:** Including this structured setup guide directly in your GitHub repository's `README.md` ensures your professor and teammates can run both your Java backend and dashboard effortlessly without friction.